Torque

Letting an AI coding agent operate real Salesforce orgs — and making it structurally unable to write to production on its own.

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Not “asks first” — denied by default, in a deterministic hook that *parses* the command instead of reading it. So `x=sf; $x data delete bulk --target-org acme-prod` never reaches the CLI, even though the gate never sees the letters `sf`.

Run `python3 bin/torque-demo` and watch two dozen real attacks fail in about 30 seconds — no org, no credentials, nothing configured. Every one of them defeated an earlier version of this code. Actual output:

```
DENIED    x=sf; $x data delete bulk --target-org acme-prod
          a shell variable assembles the command at runtime
DENIED    cat ~/.torq*/secret
          a glob that only becomes the signing secret's path once bash expands it
DENIED    : >hooks/lib.py >/tmp/z
          two redirects — bash truncates both files; the gate is the first one
DENIED    git checkout HEAD~5 -- hooks/
          restoring an older, weaker version of the gate itself
allowed   sf data query --query "SELECT Id FROM Account" -o any-org
          reads need no authorization
allowed   grep -rn 'sf data delete --target-org' .
          sf as a search string, not a command
```

The last two rows matter as much as the first four. **A guardrail that blocks real work gets switched off in week two.**

128 adversarial fixtures, each an attack that once worked

11 mutation tests that break a guard and require the attack to succeed

59 live checks run against a real org, verified by SOQL

55 defects found by adversarial review, every one now a test

> What it does *not* stop, stated here rather than in a footnote: a same-user actor running arbitrary code outside the agent's tool surface. That is a credentials boundary, not something a hook can adjudicate. Section 5 gives the specifics.

Why prompt instructions don't hold

Frontier agents can now operate Salesforce orgs directly — query, deploy, run Apex, move data — through the CLI and the official MCP server. That is enormously useful and genuinely dangerous: the same session that fixes a validation rule can, with one mistaken alias, write to a client's production org. And a client's production org is somebody's payroll, case file, or donor history.

The common answer is a sentence in a prompt: *be careful, always confirm before writing to production.* That works right up until the model is three steps into a task and pattern-matching hard — which is exactly when you need it. Torque is the layer that does not need to be obeyed.

How it holds — five layers

1 · Credentials

Connect production read-only. Torque stores no secrets; the `sf` CLI is the only credential path. This is the load-bearing layer, and the tool says so.

2 · Authorization by identity, not inference

A non-production write is allowed only when its target is on an explicit allowlist **and** classifies non-production from a live `Organization` query at the moment of the write — never from an alias, a URL, or a cached verdict. Unverifiable orgs classify as production.

3 · Deterministic gates

Two PreToolUse hooks share one expansion-aware parser. They authorize by parsing argv and default to deny, so indirection, subshell and brace grouping, wrapper runners, interpreters and here-strings, `xargs` stdin, legacy `force:data:*` verbs, and glob or `$var` path tricks are refused rather than guessed. A gate that crashes or times out denies rather than opening.

4 · Production override — deliberate, not impossible

Real client work includes deploying to production. So it is gated on a present operator, not walled off: `torque approve <org> <op> --prod` shows you the org and username, requires you to type `WRITE PRODUCTION` at a real terminal, and mints a single-use, HMAC-signed token. **The agent can request one. Through its tool surface it provably cannot create one.**

5 · Verified enforcement

Every rule is labeled hook-enforced, harness-enforced, or model-honored — and a check fails the build if a label names a hook or check that does not exist. Nothing claims more than it proves.

How it is verified

Every fixture in the suite is an attack that worked once. Four independent adversarial reviews — one of which *executed* each exploit against a live `sf` CLI rather than reading the code — broke earlier versions of these gates 55 times. Each break is now a named, runnable regression test that must deny. So 128 is not a bug count; it is the number of things you would have to invent something new to get past.

Eleven mutation tests then neuter one guard apiece and *require* the matching attack to succeed, because a check that cannot fail proves nothing. Eight run on any clone; three exercise the clean-IP scan and need a pattern list that is deliberately not published, so on your machine they report as operator-only instead of pretending to pass.

The capability run happens against a real Developer Edition org, and asserts the org state rather than the exit code: deploy a field → verify by SOQL → verify field-level security → hard-delete → confirm zero residue; a mass update with a working undo; a live Lightning render.

CLAIM	HOW IT IS CHECKED
Production is denied by default	Live org query at write time; the matrix asserts a denied write leaves the record present
The agent cannot mint approval	`torque-approve` refused without a login TTY; forged and expired tokens rejected
Destructive ops need approval	Denied without a token; the same op executes with one, and the token is consumed
The gates cannot be disabled	Redirects, glued redirects, `cd`-desync and `git checkout` at gate files all denied
A broken gate fails closed	A hung `sf` denies within the gate's wall-clock budget instead of timing out open

If a check cannot run — no browser installed, for instance — the run reports `DEGRADED`, never `PASS`. Refusing to call an unrunnable check green is the whole posture in one rule.

Where the boundary honestly sits

The gates bind the agent's **tool surface**: Bash, Edit, Write, Read, MCP. Within it they stop accidents with certainty and defeat enumerable circumvention. They do not claim to stop a same-user actor who steps outside that surface by executing arbitrary code:

- reading the signing secret by OS means the agent's gated tools never touch, or forging a login session with a purpose-built program — the filesystem and credentials trust boundary; and
- having the agent write a script and run it, so `sf` executes as a subprocess no hook ever sees.

For those the load-bearing defense is layer 1 — connect production read-only, and never leave a production org authenticated in an autonomous session. Closing the subprocess channel properly needs a PATH-level shim that classifies before `exec`; that is the v2 roadmap and it is **not built yet**. Saying so is the point: a safety claim you cannot reproduce is marketing, not security.

Run it against your own org

Prerequisites: `python3` 3.8+, the Salesforce CLI, and any non-production org — a free Developer Edition is fine.

```
python3 bin/torque-demo                    # 30s, no org, no credentials

sf org login web --alias my-dev-org
python3 bin/torque-init my-dev-org         # refuses to allowlist a production org
npm install                                # optional; only the live browser check needs it
python3 harness/validate.py --profile release --target-org my-dev-org
```

`torque-init` creates the trust anchor outside the repo and proves the gates bind before it tells you it worked. The allowlist is deliberately not shipped: which org you may write to is a decision only the person at the keyboard can make.

What it isn't

NOT THIS	THAT CATEGORY BELONGS TO
A CI/CD pipeline	Gearset, Copado
An in-org codegen IDE	Agentforce Vibes
A command library	sfdx-hardis — good prior art; Torque could sit on top of it

Torque is the operator-grade safety and validation layer those categories don't ship.

> Found a way through? That is the point — every fixture exists because a review beat an earlier version of this code. `SECURITY.md` has the disclosure path. The full validation log and audit trail are in `harness/VALIDATION.md`.